Carnegie Mellon University
Dietrich College
Information Systems

Geographic Information Systems
In-Class Exercise

Basic Earth Science: Earthquakes, Volcanoes and Tectonic Plates

In this exercise, we create maps that give us insight into some of our planet's natural seismic phenomena.

"An earthquake is a natural phenomenon characterized by the sudden shaking or trembling of the ground, caused by the movement of tectonic plates beneath the Earth's surface. These movements can occur due to various factors, including volcanic activity, the shifting of the Earths crust, or the release of stress accumulated along geological faults. The energy released during an earthquake travels through the Earth in the form of seismic waves, which is what we feel as shaking."

Earthquake magnitude is a measure of the size of an earthquake at its source. It is a logarithmic scale. For example, the amplitude of seismic waves of a magnitude 5 earthquake is 10 times as large as a magnitude 4 earthquake. A magnitude 3 to 4.9 earthquake is considered minor; 5 to 6.9 is moderate to strong; 7 to 7.9 is major; and 8 or more is an extremely powerful earthquake.

Earthquake intensity measures the strength of shaking and damage experienced at a specific location. The intensity of an earthquake at a location is a number (I - XII) that characterizes the severity of ground shaking at that location by considering the effects of the shaking on people, on built structures, and on the landscape.

THE MAP

From the home page of your CMU ArcGIS Online account, create a new map (Map option at the top). A base map should appear. We next add various layers to visualize seismic activity.

Select the 'Add layer' option and, instead of 'My content', select 'Living Atlas'. In the Search box, enter 'Earthquakes'. A list of potential layers appears. Add the layer 'Recent Earthquakes' by ESRI to your map. The layer appears on your map and the legend appears on the right of your work area. Note how the recent earthquakes are color coded. You can collapse the Layers panel for a larger view of your map. Pan and Zoom around your map to see where earthquakes occur globally. Click on a few for details.

Can you notice any patterns in the distribution of earthquakes? Of major earthquakes?

When you reach a convenient check point in your work, save your map (icon on the left; highlighted with a blue dot.) Give your map a descriptive name and add some useful tags.

To add an additional layer, select either the Add option (plus sign) or Layers / Add. Again, select 'Living Atlas' and search the term 'Volcano'. Add the layer 'SIgnificant Global Volcanic Eruptions' to your map. Close the Layers pane and note the legend on the right. Close that pane. Pan and Zoom around your map. If you open the Layers pane on the left you can toggle each layer on and off or rearrange them.

Is there any apparent relationship between the locations of recent earthquakes and volcanic activity?

Add one more layer, this time from 'ArcGIS Online' rather than 'Living Atlas'. Search 'Tectonic' and add the layer ‘Earth’s Tectonic Plates’ by ESRI Canada Education and Research. Hit the home button (lower right) to enlarge your map to the default view. You can hide the Properties pane at the right and the Layers pane at the left. Save your map.

Pan and Zoom around your map. Do you see the 'Ring of Fire'? (Some of most densely populated countries and cities on earth sit directly on the Ring of Fire. Name a few! Do you personally know any of these places?)

Finally, experiment with Base maps - the fourth icon in the left side panel. You will generally want to select base maps that highlight the features you want to demonstrate in a useful and pleasing way. (Personally, I like 'Imagery Hybrid' or 'National Geographic' for this application.)

You can see that adding layers to a map is an easy task; interpreting the underlying story is the challenge!

When you are happy with your work, save your map. When you return to ArcGIS Online, you will find it under Content.

When you are ready, we will try a filter function together.